2D Explicit FEA — Copper Panel Impact

README | MATLAB source: copper_panel_impact_fea.m | Theory: see THEORY.pdf

A self-contained MATLAB script that simulates a small steel debris sphere striking (1) a **copper (C10100) panel** and (2) the **top wall of a titanium (Ti-6Al-4V) frame**. It uses a hand-written 2D plane-stress finite element code with explicit time integration and a live von Mises stress animation. No toolboxes and no external data files are needed; the whole solver is one file.

What it does

Impact	Target	Material	Thickness	Boundary condition
1	Right panel, 280 × 20 mm	Copper C10100 (E = 115 GPa, ρ = 8960 kg/m^3, ν = 0.35)	1.32 mm	Left edge clamped
2	Frame top wall, 200 × 10 mm	Ti-6Al-4V (E = 116 GPa, ρ = 4430 kg/m^3, ν = 0.34)	10 mm	Both ends clamped

Projectile: steel sphere, radius 2.5 mm, density 7850 kg/m^3 (mass ≈ 0.514 g), fired straight down at 1000 m/s (kinetic energy ≈ 257 J). Each target is hit separately at mid-span.

Outputs:

- Live animation of the deformed meshes coloured by (normalised) von Mises stress.
- Time histories of maximum nodal displacement for the copper panel and the titanium frame.
- Console summary of peak deflection for both parts.

Requirements

- **MATLAB R2020b or newer** (uses turbo, yline with labels, gobjects).
- A desktop MATLAB session with graphics. Not tested in GNU Octave and not expected to work there.
- Save the script as **UTF-8**. It prints μ, — and ✓, which may look garbled in older Windows consoles but does not affect results.

Quick start

```
>> copper_panel_impact_fea
```

Expected console output at start-up:

```
============================================
  DUAL IMPACT FEA – COPPER PANEL
============================================
dt       = 312.675 ns
T_end    = 650 μs
Steps    = 2079

Running Copper panel impact simulation...
  IMPACT #1 (COPPER Panel) at t=... μs
  IMPACT #2 (Ti Frame) at t=... μs
```

Contact begins roughly 2 μs in, because the debris starts 2 mm above each surface. At the end the script prints the maximum deflection of the copper panel and the titanium frame.

Configurable parameters

All inputs are at the top of the script.

Variable	Meaning	Default
E_cu, nu_cu, rho_cu	Copper properties	115 GPa, 0.35, 8960 kg/m^3
E_ti, nu_ti, rho_ti	Titanium properties	116 GPa, 0.34, 4430 kg/m^3
R_d, rho_d	Debris radius, density	2.5 mm, 7850 kg/m^3
dVy1, dVy2	Debris velocity (m/s, negative = downward)	−1000
panel_L, panel_H, panel_th_cu	Copper panel length, height, thickness	280 mm, 20 mm, 1.32 mm
frame_outer, frame_wall, frame_th_ti	Frame size, wall width, thickness	200 mm, 10 mm, 10 mm
NX_p, NY_p	Copper mesh divisions	80 × 10
NX_f, NY_f	Titanium wall mesh divisions	50 × 5
k_pen_cu, k_pen_f	Contact penalty stiffness (E·R_d·32)	≈ 9.2e9 and 9.3e9 N/m
T_end	Simulated time	650 µs

The time step is computed automatically from the smallest element size and the fastest wave speed (0.8 safety factor). If you refine the mesh, the step shrinks by itself and run time grows accordingly.

Code layout

Section	Purpose
Materials / Geometry	Properties, wave speeds, debris mass, dimensions
D-matrices	Plane-stress constitutive matrices
Mesh 1 / Mesh 2	Structured quad meshes, B-matrices, lumped masses, boundary nodes
Timestep	CFL-based dt and step count
State vectors / Debris	Displacements, velocities, initial debris positions
Figure setup	Axes, plots, colour map, static drawing of the structure
Main time loop	Internal forces, contact forces, explicit update, debris motion, history, animation
Final results	Console summary

Model assumptions and limitations

Please read these before using the numbers for anything beyond a demonstration.

1. **Linear elastic only.** No plasticity, strain-rate hardening, damage or erosion. At 1 km/s on a 1.32 mm copper panel, real behaviour would involve yielding and possibly perforation, so the reported deflections are elastic-model values, not predictions of physical damage.
2. **Plane stress, 2D.** Each target is a flat sheet in its own plane; out-of-plane bending is not modelled. The debris is a 2D disk for contact but carries the mass of a 3D sphere.
3. **One-point integration without hourglass control.** Strains are evaluated at the element centre. Zero-energy (hourglass) modes are possible on coarse meshes.
4. **Penalty contact.** Panel nodes inside the debris radius are pushed out with a linear spring, so results depend on k_pen_*.
5. **Targets are independent.** The copper panel and titanium frame are separate meshes and are not connected to each other.
6. **Pseudo-fixed supports.** Clamped nodes are fixed and also given 200× mass.

7. **Explicit integration** (semi-implicit Euler, leapfrog form). Stability relies on the CFL step; if you change the mesh aspect ratio or materials drastically, reduce the 0.80 factor.
8. **Stress colours are normalised** to the current maximum, so the colour bar shows relative, not absolute, stress.

Known cleanup items

These do not affect results but are worth tidying before publishing:

- The "Calibration Error" plot (ax3, h_error, h_dot3) is created but never updated.
- htxt_phase, cu_col, imp_x1/y1, imp_x2/y2 and wave_r1/r2 are set but not used for display.
- The commented-out line before the final fprintf has mismatched quotes; it is harmless as a comment but worth fixing.
- The comment says "Velocity-Verlet"; the update is actually semi-implicit (symplectic) Euler, equivalent to leapfrog central differences.
- The element loops are not vectorised. Vectorising over elements, or updating patch colours less often, would speed up long runs noticeably.

Repository contents

```
copper_panel_impact_fea.m          the simulation
copper_panel_impact_fea_code.pdf   printable listing of the source
README.pdf                         this document
THEORY.pdf                         equations and method
```

License

Add the license of your choice (for example MIT) as a LICENSE file before publishing.